AEGIS-TWIN

Forensic Incident Report

SEVERITY: CRITICAL

evice Name	Coolant Pump	**Device ID**	DEV-008
ector / Zone	8	**Timestamp**	09 May 2026 16:57:53 UTC
rust Score	15.7 / 100	**Reconstruction MSE**	0.48288
ensen-Shannon Div	0.37707	**Attack Pattern**	Data Exfiltration — large payloads with elevate

Incident Signature (SHA-256)	FE81C5C5C830051DA97F0F4B26EE0CB7483122833D37B06ED1AD404DDD21F204

1 — Executive Summary

A **CRITICAL** severity anomaly was detected. The device exhibits behaviour strongly inconsistent with its established baseline and must be treated as actively compromised. Immediate isolation and incident response are required.

The Aegis-Twin digital twin engine, using an LSTM autoencoder, observed device **Coolant Pump** (ID: **DEV-008**) in sector **8** at **09 May 2026 16:57:53 UTC**. The trust score fell to **15.7 / 100** and reconstruction error rose to **0.48288**, triggering automated forensic collection and report generation.

2 — Digital Twin Analysis Metrics

The following metrics were recorded by the autoencoder at the time of detection. Elevated MSE and JSD values indicate that the observed traffic distribution diverges significantly from the trained baseline.

Metric	Observed Value	Reference / Threshold
Trust Score	15.7 / 100	*Healthy range: ≥ 75*
Reconstruction Error (MSE)	0.482884	*Threshold: 0.20 (medium) / 0.35 (high) / 0.50 (critical)*
Jensen-Shannon Divergence	0.377070	*Threshold: 0.30 / 0.50 / 0.70*
Derived Severity	**CRITICAL**	*Computed from the above three metrics*

3 — Top Anomaly Contributing Features

The features below showed the largest deviation between the observed traffic and the trained baseline. Higher delta values indicate greater anomaly contribution.

Feature	Baseline	Observed	Delta	% Change	Direction
Inter-Arrival Time (IAT)	0.4000	1.0000	0.6000	150.0%	↑ Increased
Payload Entropy	0.4000	1.0000	0.6000	150.0%	↑ Increased
Flow Symmetry	0.6000	0.0000	0.6000	100.0%	↓ Decreased

4 — Digital Evidence Snapshot

Raw feature vectors captured at detection time for archival and chain-of-custody purposes.

Feature	Baseline	Current	Abs. Delta
Packet Size	0.50000	0.00000	0.50000
Inter-Arrival Time (IAT)	0.40000	1.00000	0.60000
Payload Entropy	0.40000	1.00000	0.60000
Flow Symmetry	0.60000	0.00000	0.60000

5 — Identified Attack Pattern

Classified pattern: Data Exfiltration — large payloads with elevated entropy

The classification is derived from heuristic analysis of feature deltas and packet inter-arrival timing distributions. It should be treated as an investigative hypothesis and validated by a human analyst.

6 — Packet Traffic Summary

#	Source	Destination	Protocol	Size (B)	IAT (s)
1	—	—	—	—	0.0285
2	—	—	—	—	0.0428
3	—	—	—	—	0.0195
4	—	—	—	—	0.0037
5	—	—	—	—	0.045
6	—	—	—	—	0.0196
7	—	—	—	—	0.0132
8	—	—	—	—	0.013
9	—	—	—	—	0.0116
10	—	—	—	—	0.0457
11	—	—	—	—	0.0472
12	—	—	—	—	0.0302

7 — Behavioural Timeline

No threat log entries were recorded at the time of report generation.

8 — Risk Assessment

This incident is rated **CRITICAL**. The combination of very low trust score, high reconstruction error, and significant distribution divergence indicates a strong likelihood of active exploitation or data exfiltration. The device should be treated as **fully compromised** until cleared by forensic investigation.

9 — Recommended Remediation Actions

Priority	Remediation Action
Immediate	Isolate the device from the network (quarantine VLAN or physical disconnect).

CRITICAL

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Immediate	Preserve volatile memory: capture a RAM image before any reboot.
Short-term	Collect full packet captures and system/application logs for post-incident analysis.
Short-term	Perform a firmware integrity check and compare against known-good hash.
Short-term	Rotate all credentials, API keys, and revoke active sessions associated with the device.
Long-term	Update IDS/IPS signatures to detect similar behavioural patterns.
Long-term	Review and tighten network segmentation and access-control lists for this sector.
Long-term	Schedule a post-incident review and update the Aegis-Twin baseline after remediation.